# Mapping the U.S. Healthcare System with Public Data

*Data-engineering experiment + AI-agent implementation handoff*

Core question: There are millions of healthcare providers and thousands of hospitals. Why can it still be difficult to get an appointment, and what does the healthcare system actually look like when modeled as data?

Primary case study: build the national data model first, then produce a Connecticut-focused analysis as the concrete story. The project should be able to expand nationally after the pipeline is proven.

Recommended stack: Python + DuckDB/Polars for ingestion and profiling, BigQuery for the analytical warehouse, dbt for transformations/tests, and a lightweight visualization layer such as Looker Studio, Streamlit, or Observable. The agent may choose an equivalent stack if it keeps the same data contracts.

## 1. What this experiment is trying to prove

Do not begin by assuming that appointment difficulty is caused by a simple shortage of doctors. Test competing explanations.

| Hypothesis | What the data can test | Likely evidence |
|---|---|---|
| H1: There are not enough providers. | Provider density by specialty and geography. | Providers per 100k; specialist scarcity. |
| H2: Providers are badly distributed. | Compare provider density with population density. | Large geographic variance. |
| H3: The usable provider pool is smaller than the total pool. | Compare all NPIs with Medicare-enrolled clinicians, specialty/location filters, and network data. | Large drop after filters. |
| H4: Some providers/groups carry very high utilization. | Medicare utilization by NPI and specialty. | High concentration of services. |
| H5: Hospital/medical-group concentration matters. | Provider-to-facility and provider-to-organization relationships. | Dense affiliation clusters. |
| H6: Insurance networks are a major access filter. | Marketplace network metadata + selected machine-readable files. | Network-specific provider availability differs. |

Important: public data does not directly measure appointment wait time for every clinician. The project must not turn a proxy into a fact. Instead, build an 'access pressure' analysis from observable supply, geography, network participation where available, and utilization.

## 2. Public datasets to use

Use official sources as the canonical inputs. Record source URL, retrieval timestamp, dataset version, file checksum, and transformation version in an ingestion manifest.

| Priority | Dataset | Purpose | Key identifiers / fields | Source |
|---|---|---|---|---|
| P0 | NPPES NPI Downloadable File V2 | Master provider/entity/location universe. | NPI, Entity Type, Legal Business Name, taxonomy, practice address, enumeration/deactivation dates. | https://download.cms.gov/nppes/NPI_Files.html |

| | | | | |
|---|---|---|---|---|
| P0 | NPPES Practice Location Reference File | Additional Type 1/Type 2 practice locations. | NPI + location fields. | https://download.cms.gov/nppes/NPI_Files.html |
| P0 | CMS Doctors & Clinicians National Downloadable File | Medicare-enrolled clinician/group/address records. | NPI, PAC IDs, specialty, facility name, organization PAC, address, assignment. | https://data.cms.gov/provider-data/dataset/mj5m-pzi6 |
| P0 | CMS Facility Affiliation Data | Clinician-to-facility relationships. | NPI + facility affiliation fields; inspect current dictionary. | https://data.cms.gov/provider-data/dataset/27ea-46a8 |
| P0 | CMS Hospital General Information | Hospital universe, address, type, ownership/rating fields. | Facility/CCN identifier, name, address, state, type. | https://data.cms.gov/provider-data/dataset/xubh-q36u |
| P1 | CMS Medicare Physician & Other Practitioners - by Provider | Annual Medicare utilization/payment profile by NPI. | NPI, services, payments, beneficiaries, charges and related measures. | https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners/medicare-physician-other-practitioners-by-provider |
| P1 | CMS Medicare Physician & Other Practitioners - by Provider and Service | Procedure/service-level utilization. | Rendering NPI, HCPCS, place of service, utilization/payment fields. | https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners/medicare-physician-other-practitioners-by-provider-and-service |
| P1 | Census ACS 5-year | Population and socioeconomic denominator. | ZCTA/county population and demographic estimates. | https://www.census.gov/data/developers/data-sets/acs-5year.html |
| P1 | Census TIGER/Line | Geographic boundaries and spatial joins. | ZCTA/county/tract geometries. | https://www.census.gov/geographies/mapping-files.html |
| P1 | CMS Marketplace Exchange PUFs | Plan/service-area/network metadata. | Plan ID, issuer, county/ZIP service area, network URL, machine-readable URL. | https://www.cms.gov/marketplace/resources/data/public-use-files |
| P2 | Transparency in Coverage MRFs | Deep network/provider/rate exploration for selected plans. | Plan/network references and provider identifiers where exposed by file. | https://www.cms.gov/priorities/healthplan-price-transparency/overview/use-pricing-information-published-under-transparency-coverage-final-rule |

Current CMS notes: the NPPES page says Version 2 is the recommended downloadable format and, as of March 3, 2026, Version 1 is no longer supported. The September 14, 2026 monthly V2 file is listed at about 1.1 GB compressed. NPPES also supplies other-name, practice-location, and endpoint reference files. CMS warns that having an NPI does not validate licensure or credentialing. cite turn3search0

The CMS Doctors & Clinicians National Downloadable File currently has 3,388,628 rows and 31 columns and is unique at clinician/enrollment/group/address level rather than one row per person. That means COUNT(*) must never be interpreted as number of doctors. cite turn1search1 turn1search9

CMS's Facility Affiliation Data currently contains more than 2.2 million rows and exposes an API/download. cite turn1search4

CMS Hospital General Information currently contains 5,419 rows and 38 columns and provides an API/download. cite turn0search6

CMS's Medicare Physician & Other Practitioners by Provider dataset is annual; the latest available dataset is 2024 and contains provider-level utilization/payment information keyed by NPI. cite turn2search3 turn2search4

CMS's Exchange PUF collection includes Plan, Service Area, Network, and Machine-readable URL files. This gives a practical route into insurance-network analysis without pretending that every commercial network is directly available as a clean national table. cite turn2search1

Transparency in Coverage machine-readable files are much larger and more complex. CMS explicitly describes their size and complexity and says they are intended for third-party processing. Treat them as a Phase 2/advanced ingestion project, not the first pipeline. cite turn2search0 turn2search10

## 3. Recommended project phases

| Phase | Scope | Exit criterion |
|---|---|---|
| 0 | Reproducible environment + source manifest. | One command records versions/checksums and creates raw/ directories. |
| 1 | NPPES + Census geography. | Can count unique NPIs, classify entities, and map locations. |
| 2 | CMS Doctors & Clinicians + facility affiliations. | Can connect clinicians to Medicare enrollment records, groups, addresses, and facilities. |
| 3 | Hospitals + Medicare utilization. | Can measure hospital/provider supply and utilization proxies. |
| 4 | Marketplace network metadata. | Can compare plan/service-area/network metadata for selected states/plans. |
| 5 | Connecticut case study. | Dashboard and SQL results are reproducible for CT. |
| 6 | National expansion. | Same models run across all states without special-case logic. |
| 7 | Story packaging. | Charts, methodology, caveats, and Medium-ready narrative generated from versioned outputs. |

## 4. Warehouse architecture

```
raw
├── nppes
│   ├── npi_monthly_v2
│   ├── practice_locations
│   ├── other_names
│   └── endpoints
├── cms
│   ├── clinicians_national
│   ├── facility_affiliation
│   ├── hospital_general
│   ├── mp_provider
│   └── mp_provider_service
├── census
│   ├── acs
```

```
│   └── tiger_zcta
└── marketplace
    ├── plan_puf
    ├── service_area_puf
    ├── network_puf
    └── mr_url_puf

staging
├── stg_npi
├── stg_npi_location
├── stg_clinician
├── stg_affiliation
├── stg_hospital
├── stg_utilization
├── stg_population
└── stg_marketplace

core
├── dim_provider
├── dim_organization
├── dim_location
├── dim_facility
├── dim_specialty
├── dim_plan
├── bridge_provider_location
├── bridge_provider_group
├── bridge_provider_facility
├── bridge_plan_service_area
└── fact_provider_utilization

mart
├── mart_provider_supply
├── mart_specialty_supply
├── mart_geographic_access
├── mart_provider_network
├── mart_provider_concentration
├── mart_facility_ecosystem
└── mart_access_pressure
```

## 5. Core warehouse schema

Use stable surrogate keys internally, but preserve source identifiers. NPI is the natural external identifier for providers; CMS facility identifiers/CCNs should remain separate from NPI because NPI is not a facility identifier and is 'intelligence-free.'

| Table | Grain | Important columns |
|---|---|---|
| dim_provider | One current provider/entity per NPI | provider_sk, npi, entity_type, first_name, last_name, legal_name, enumeration_date, deactivation_date, primary_taxonomy, taxonomy_list, source_updated_at |
| dim_organization | One organization NPI | organization_sk, npi, legal_business_name, dba_name, taxonomy, address_sk |
| dim_location | One normalized location | location_sk, address1, address2, city, state, zip5, zip4, county_fips, lat, lon, geocode_quality |
| dim_specialty | One taxonomy/specialty | specialty_sk, taxonomy_code, taxonomy_description, specialty_group, is_primary |
| dim_facility | One facility | facility_sk, ccn, facility_name, facility_type, address_sk, state, source |

| | | |
|---|---|---|
| dim_plan | One marketplace plan/version | plan_sk, plan_id, issuer_id, state, plan_year, plan_type |
| bridge_provider_location | Provider x location | provider_sk, location_sk, source, is_primary, effective_date, end_date |
| bridge_provider_group | Clinician x group/enrollment | provider_sk, org_pac_id, group_assignment, enrollment_id, source_snapshot |
| bridge_provider_facility | Clinician x facility affiliation | provider_sk, facility_sk or facility_match_key, affiliation_type, source_snapshot |
| bridge_plan_service_area | Plan x geography | plan_sk, state, county_fips, zip5, service_area_source |
| fact_provider_utilization | NPI x year | provider_sk, year, services, beneficiaries, allowed_amount, Medicare_payment, submitted_charges, source |
| fact_provider_service | NPI x HCPCS x year | provider_sk, year, hcpcs, place_of_service, services, beneficiaries, allowed_amount, payment |
| fact_population | Geography x year | geo_type, geo_id, year, population, source |

## 6. Entity resolution and joins

The most important engineering rule: do not infer relationships from names when a source identifier exists. Keep source IDs and create explicit match-quality fields when a relationship must be inferred.

| Relationship | Preferred join | Fallback | Confidence |
|---|---|---|---|
| NPPES provider → CMS clinician | NPI = NPI | None | High |
| CMS clinician → Medicare utilization | NPI = Rndrng_NPI / NPI | None | High |
| CMS clinician → facility affiliation | NPI = NPI | None | High for clinician side |
| Hospital → facility affiliation | Use facility identifier/CCN if exposed and stable | Normalized facility name + address | High/medium |
| Provider → organization | Use CMS org PAC/group fields where supplied | Organization NPI/name/address only if source supports it | Medium; document |
| Location → ZIP | ZIP5 | Address geocoding | High for ZIP, variable for coordinates |
| ZIP → county | Census ZCTA/crosswalk/spatial join | Address geocode | High |
| Plan → service area | Plan ID + PUF service-area keys | None | High |
| Plan → provider network | Use network/MRF identifiers where available | Manual selected-plan validation | Variable |

A crucial distinction: a Type 2 NPI is an organization, but the existence of an organization NPI does not automatically prove that an individual clinician is employed by that organization. CMS's clinician file has group/PAC fields that should be preferred for the clinician-to-group relationship.

## 7. Canonical transformations

1. Normalize NPI to a 10-character string. Never cast it to an integer.
2. Split NPPES Type 1 individuals and Type 2 organizations into separate logical entities.

3. Explode NPPES taxonomy fields into a provider-specialty bridge rather than keeping only one taxonomy.
4. Explode primary and reference practice locations into a provider-location bridge.
5. Normalize ZIP to ZIP5 and retain the original raw ZIP field.
6. Attach county/state using Census geography. Keep a geography_match_method column.
7. Deduplicate clinician rows at the correct analytical grain; the CMS clinician file is not person-grain.
8. Use NPI as the cross-source key for provider-level utilization.
9. Use explicit facility identifiers when available; do not equate facility_name with a unique facility.
10. Create a source_snapshot_date on every core bridge/fact table.
11. Maintain row-count and uniqueness tests for every source and transformation.
12. Create a data_quality table containing source, snapshot, rows_loaded, rows_rejected, null_rate, duplicate_rate, and checksum.

# 8. SQL analyses — 15 experiments

The examples below use BigQuery-style SQL. Replace project.dataset names with the actual environment. The agent should implement each query as a versioned model or analysis file and save the output used in the article.

## 8.1 How many unique NPIs exist by entity type?

```
SELECT
  entity_type_code,
  COUNT(DISTINCT npi) AS unique_npis
FROM core.dim_provider
GROUP BY 1
ORDER BY unique_npis DESC;
```

Story: NPI count is a provider/entity enumeration count, not a count of practicing doctors.

## 8.2 How many individual clinicians by specialty?

```
SELECT
  specialty_group,
  COUNT(DISTINCT provider_sk) AS clinicians
FROM core.dim_provider_specialty
WHERE entity_type_code = '1'
GROUP BY 1
ORDER BY clinicians DESC;
```

## 8.3 Primary-care supply per 100,000 population

```
WITH supply AS (
  SELECT state,
         COUNT(DISTINCT provider_sk) AS providers
  FROM mart.mart_provider_supply
  WHERE specialty_group = 'Primary Care'
  GROUP BY state
)
SELECT
  s.state,
  s.providers,
  p.population,
  SAFE_DIVIDE(s.providers * 100000, p.population) AS providers_per_100k
FROM supply s
JOIN mart.state_population p USING (state)
ORDER BY providers_per_100k DESC;
```

## 8.4 Specialist density by county

```
SELECT
  county_fips,
  specialty_group,
```

```
  COUNT(DISTINCT provider_sk) AS providers,
  MAX(population) AS population,
  SAFE_DIVIDE(COUNT(DISTINCT provider_sk) * 100000, MAX(population)) AS providers_per_100k
FROM mart.mart_specialty_supply
GROUP BY county_fips, specialty_group
HAVING MAX(population) > 0
ORDER BY providers_per_100k;
```

## 8.5 Where are providers concentrated?

```
SELECT
  state,
  COUNT(DISTINCT provider_sk) AS providers,
  COUNT(DISTINCT location_sk) AS locations,
  COUNT(DISTINCT organization_sk) AS organizations
FROM mart.mart_provider_supply
GROUP BY state
ORDER BY providers DESC;
```

## 8.6 How many providers have multiple practice locations?

```
SELECT
  COUNTIF(location_count = 1) AS one_location,
  COUNTIF(location_count BETWEEN 2 AND 5) AS two_to_five,
  COUNTIF(location_count > 5) AS six_plus
FROM (
  SELECT provider_sk, COUNT(DISTINCT location_sk) AS location_count
  FROM core.bridge_provider_location
  GROUP BY provider_sk
);
```

## 8.7 How many CMS clinician records collapse to one NPI?

```
SELECT
  COUNT(*) AS clinician_rows,
  COUNT(DISTINCT npi) AS unique_npis,
  SAFE_DIVIDE(COUNT(*), COUNT(DISTINCT npi)) AS rows_per_npi
FROM staging.stg_clinician;
```

This is a key engineering/storytelling chart: it demonstrates why naïve row counts produce misleading 'number of doctors' claims.

## 8.8 Providers per facility and facility concentration

```
SELECT
  facility_id,
  facility_name,
  COUNT(DISTINCT npi) AS affiliated_clinicians
FROM core.bridge_provider_facility
GROUP BY facility_id, facility_name
ORDER BY affiliated_clinicians DESC;
```

## 8.9 Provider concentration by organization

```
SELECT
  org_pac_id,
  COUNT(DISTINCT provider_sk) AS clinicians
FROM core.bridge_provider_group
WHERE org_pac_id IS NOT NULL
GROUP BY org_pac_id
ORDER BY clinicians DESC;
```

## 8.10 Medicare utilization per clinician

```
SELECT
  specialty_group,
```

```
  COUNT(DISTINCT provider_sk) AS clinicians,
  SUM(services) AS services,
  SAFE_DIVIDE(SUM(services), COUNT(DISTINCT provider_sk)) AS services_per_clinician
FROM mart.mart_provider_utilization
GROUP BY specialty_group
ORDER BY services_per_clinician DESC;
```

## 8.11 Utilization concentration — top 1%, 5%, 10%

```
WITH ranked AS (
  SELECT
    provider_sk,
    services,
    PERCENT_RANK() OVER (ORDER BY services DESC) AS pct
  FROM mart.mart_provider_utilization
)
SELECT
  SUM(IF(pct <= 0.01, services, 0)) / SUM(services) AS top_1pct_share,
  SUM(IF(pct <= 0.05, services, 0)) / SUM(services) AS top_5pct_share,
  SUM(IF(pct <= 0.10, services, 0)) / SUM(services) AS top_10pct_share
FROM ranked;
```

## 8.12 Providers per 100k vs utilization

```
SELECT
  county_fips,
  providers_per_100k,
  services_per_provider
FROM mart.mart_geographic_access
WHERE providers_per_100k IS NOT NULL
  AND services_per_provider IS NOT NULL;
```

Visualize as a scatter plot. This does not prove appointment availability, but it can identify places where supply and observed Medicare utilization differ.

## 8.13 Marketplace plan service-area breadth

```
SELECT
  issuer_id,
  plan_id,
  plan_year,
  COUNT(DISTINCT zip5) AS zips,
  COUNT(DISTINCT county_fips) AS counties
FROM core.bridge_plan_service_area
GROUP BY issuer_id, plan_id, plan_year
ORDER BY counties DESC, zips DESC;
```

## 8.14 Network metadata coverage

```
SELECT
  state,
  COUNT(DISTINCT plan_id) AS plans,
  COUNTIF(network_url IS NOT NULL) AS plans_with_network_url,
  COUNTIF(mrf_url IS NOT NULL) AS plans_with_mrf_url
FROM mart.mart_plan_network_metadata
GROUP BY state
ORDER BY state;
```

## 8.15 The 'accessible pool' funnel

```
SELECT
  COUNT(DISTINCT provider_sk) AS all_providers,
  COUNT(DISTINCT IF(is_target_specialty, provider_sk, NULL)) AS specialty_providers,
  COUNT(DISTINCT IF(in_target_geography, provider_sk, NULL)) AS geographically_relevant,
  COUNT(DISTINCT IF(medicare_enrolled, provider_sk, NULL)) AS medicare_enrolled,
```

```
  COUNT(DISTINCT IF(network_match, provider_sk, NULL)) AS network_matched
FROM mart.mart_access_pressure;
```

This should become the centerpiece only if the network_match step is based on actual provider-level network data. If the network data is incomplete, label the output as a selected-plan case study rather than a national accessible-provider count.

## 9. Visualizations that should drive the Medium story

| # | Visualization | Purpose | Why it matters |
|---|---|---|---|
| 1 | Provider-count funnel | NPI → individual → specialty → location → Medicare-enrolled → selected network | Shows why headline provider counts can be misleading. |
| 2 | U.S./CT provider density map | Providers per 100k by county | Makes geographic inequality visible. |
| 3 | Specialty scarcity map | Primary care, cardiology, dermatology, psychiatry, orthopedics, etc. | Shows that 'doctor shortage' is specialty-specific. |
| 4 | Provider → group → facility network graph | Nodes and edges for one metro area | Explains organizations and affiliations visually. |
| 5 | Top health-system/provider-group concentration | Horizontal bar chart | Shows whether local care is concentrated in a few organizations. |
| 6 | Provider supply vs Medicare utilization scatter | x=supply/100k, y=services/provider | Tests whether low supply areas also show higher observed utilization. |
| 7 | Hospital ecosystem map | Hospitals + affiliated clinicians + groups | Turns 'a hospital' into a connected ecosystem. |
| 8 | Plan service-area comparison | Counties/ZIPs covered by selected plans | Explains that insurance is also a geographic network. |
| 9 | Selected-plan network funnel | All clinicians → specialty → geography → network | Closest public-data approximation to the appointment-access question. |
| 10 | One synthetic patient journey | PCP → specialist → imaging → hospital → pharmacy | Humanizes the graph without using real patient data. |

## 10. The central Medium narrative

Suggested title: 'I Tried to Map the U.S. Healthcare System with Public Data. Here's Why Getting an Appointment Is So Hard.'

Alternative: 'There Are Millions of Healthcare Providers. So Why Is Getting an Appointment Still So Hard?'

Opening: start with the contradiction. The public can count millions of NPIs and thousands of hospitals, yet a patient does not experience the healthcare system as a national pool of providers.

Act 1 — Count the doctors. Explain NPI/entity counts and why rows are not people.

Act 2 — Put them on a map. Show geographic and specialty differences.

Act 3 — Connect them to groups and hospitals. Reveal the healthcare graph.

Act 4 — Add Medicare utilization. Show that provider presence and provider capacity are not the same measurement.

Act 5 — Add insurance network metadata for a selected market/plan. Show the accessible pool shrinking, but only where the data supports it.

Conclusion: the useful mental model is not 'How many doctors exist?' It is 'How many appropriate providers are geographically reachable, connected to the right organizations, participating in the relevant network, and apparently carrying available capacity?' Public data can measure pieces of that question, but not every piece.

## 11. What NOT to claim

- Do not say NPPES contains the number of licensed doctors. CMS explicitly says NPI issuance does not validate licensure or credentialing.
- Do not count CMS clinician rows as people. The file is at clinician/enrollment/group/address grain.
- Do not claim Medicare utilization equals total healthcare utilization. It covers Original Medicare fee-for-service Part B data.
- Do not infer that a provider is accepting new patients from an NPI or Medicare enrollment record.
- Do not infer appointment wait time from utilization alone.
- Do not assume a hospital name is a unique corporate entity.
- Do not claim a provider is in-network for a commercial plan unless the provider-plan relationship is actually observed in the selected network data.
- Do not combine datasets from different years without exposing the year mismatch.
- Do not use patient-level or protected health information. This experiment should use public aggregate/provider/facility data only.

## 12. Data quality tests

| Test | Expectation |
|---|---|
| NPI format | Exactly 10 digits; retain as STRING. |
| NPI uniqueness | One row in dim_provider per NPI after selecting the current source record. |
| Entity type | 1 = individual; 2 = organization in NPPES logic. |
| ZIP format | ZIP5 is 5 characters; retain original value separately. |
| Provider utilization join | No unexpected many-to-many explosion on NPI. |
| Facility join | Track exact-ID vs name/address inferred matches separately. |
| Population join | Every reported geography has a source population for the selected year. |
| Snapshot consistency | Every analysis records source years/snapshot dates. |
| Row reconciliation | Raw rows = accepted + rejected; rejected rows have a reason. |
| Network matching | Every network match has source URL/file and match method. |

## 13. Suggested repository structure

```
us-healthcare-data/
├── README.md
├── AGENT_HANDOFF.md
├── pyproject.toml
├── .env.example
├── Makefile
├── config/
│   ├── sources.yml
│   ├── geography.yml
│   └── analysis.yml
├── ingestion/
│   ├── nppes/
│   ├── cms/
```

```
│   ├── census/
│   └── marketplace/
├── dbt/
│   ├── staging/
│   ├── core/
│   └── marts/
├── sql/
│   ├── 01_provider_counts.sql
│   ├── 02_specialty_supply.sql
│   ├── ...
│   └── 15_access_funnel.sql
├── tests/
├── notebooks/
│   ├── 01_profile_sources.ipynb
│   ├── 02_entity_resolution.ipynb
│   └── 03_story_validation.ipynb
├── dashboards/
├── docs/
│   ├── data_dictionary.md
│   ├── methodology.md
│   └── story_outline.md
└── outputs/
    ├── tables/
    ├── charts/
    └── maps/
```

# 14. AI-agent build instructions

Paste the following section into an AI coding agent as the implementation contract.

## AI AGENT SYSTEM BRIEF

```
You are the lead data engineer for a reproducible public-data investigation of the U.S. healthcare system.

GOAL
Build a production-quality analytical pipeline that explains why the number of healthcare providers is not
equivalent to healthcare access.

PRIMARY QUESTION
There are millions of healthcare providers and thousands of hospitals. How does the observable provider ecosystem
differ from the provider pool a patient can realistically use?

SCOPE
1. Build the provider/entity/location backbone from NPPES.
2. Add CMS Doctors & Clinicians data.
3. Add CMS Facility Affiliation data.
4. Add CMS Hospital General Information.
5. Add Medicare Physician & Other Practitioners utilization.
6. Add Census population/geography.
7. Add Marketplace Exchange PUF network/service-area metadata.
8. Implement a selected-plan network experiment only when provider-level matching is defensible.
9. Produce Connecticut results first.
10. Make the pipeline parameterized so it can run nationally.

ENGINEERING PRINCIPLES
- Prefer official public sources.
- Never fabricate fields, URLs, identifiers, or relationships.
- Preserve raw data unchanged.
- Record source URL, retrieval timestamp, source version/date, file checksum, row count, and schema.
- Treat NPI as STRING.
- Never use COUNT(*) as a person count.
- Keep source grain explicit.
- Never infer a provider relationship from a name if a source identifier exists.
```

- Track match method and confidence for inferred relationships.
- Do not use patient-level PHI.
- Do not scrape websites unless a source is explicitly approved and its terms allow it.
- Make every analysis reproducible from versioned SQL.
- Every published number must have a source snapshot and methodology note.

SOURCE REGISTRY
NPPES: https://download.cms.gov/nppes/NPI_Files.html
CMS Doctors & Clinicians: https://data.cms.gov/provider-data/dataset/mj5m-pzi6
CMS Facility Affiliations: https://data.cms.gov/provider-data/dataset/27ea-46a8
CMS Hospital General Information: https://data.cms.gov/provider-data/dataset/xubh-q36u
CMS Medicare Physician by Provider: https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners/medicare-physician-other-practitioners-by-provider
CMS Medicare Physician by Provider and Service: https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners/medicare-physician-other-practitioners-by-provider-and-service
CMS Exchange PUFs: https://www.cms.gov/marketplace/resources/data/public-use-files
CMS Transparency in Coverage: https://www.cms.gov/priorities/healthplan-price-transparency/overview/use-pricing-information-published-under-transparency-coverage-final-rule
Census ACS: https://www.census.gov/data/developers/data-sets/acs-5year.html
Census TIGER/Line: https://www.census.gov/geographies/mapping-files.html

IMPLEMENTATION ORDER
Phase 1: source manifest, NPPES ingestion, Census geography.
Phase 2: CMS clinician and affiliation ingestion.
Phase 3: hospital and utilization ingestion.
Phase 4: analytical marts.
Phase 5: Connecticut dashboard.
Phase 6: Marketplace network metadata.
Phase 7: selected-plan network experiment.
Phase 8: national expansion and story outputs.

REQUIRED TABLES
dim_provider
dim_organization
dim_location
dim_specialty
dim_facility
dim_plan
bridge_provider_location
bridge_provider_group
bridge_provider_facility
bridge_plan_service_area
fact_provider_utilization
fact_provider_service
fact_population

REQUIRED MARTS
mart_provider_supply
mart_specialty_supply
mart_geographic_access
mart_provider_network
mart_provider_concentration
mart_facility_ecosystem
mart_access_pressure

REQUIRED ANALYSES
Implement the 15 analyses in this document. Each query must be stored in source control and have a short methodology comment.

REQUIRED OUTPUTS
- source_manifest.csv
- data_quality_report.csv
- provider_counts.csv

```
- specialty_supply.csv
- county_supply.csv
- facility_affiliations.csv
- provider_concentration.csv
- utilization_summary.csv
- network_metadata.csv
- access_funnel.csv
- charts/
- maps/
- methodology.md
- story_outline.md

ACCEPTANCE CRITERIA
1. A fresh environment can run the pipeline from documented commands.
2. Raw files are not modified in place.
3. All datasets have source metadata.
4. NPI joins do not create unexplained many-to-many explosions.
5. Provider counts distinguish unique NPI from CMS clinician rows.
6. Connecticut can be selected through configuration rather than hard-coded SQL.
7. Every chart is generated from a versioned query/output.
8. Every published claim can be traced to a dataset, snapshot, and SQL model.
9. The final README explains known limitations.
10. The agent must stop and ask for human review when a required relationship cannot be supported by a public
identifier.

DO NOT
- invent appointment wait times
- invent insurance network participation
- claim that NPI = licensed doctor
- claim Medicare utilization = all utilization
- silently deduplicate source records
- silently mix years
- publish personal or patient-level information

FIRST TASK
Before writing transformation code, create:
1. source_manifest.yml
2. proposed schemas
3. ingestion plan
4. data-quality checklist
5. exact source files/endpoints to be used
Then wait for review before beginning large downloads.
```

## 15. Definition of done for the article

| Requirement | Done when |
|---|---|
| Data | At least 5 official datasets are integrated with documented keys. |
| Engineering | Pipeline is reproducible and incremental where practical. |
| Entity resolution | Provider, organization, location and facility relationships have explicit source/match logic. |
| Analysis | At least 10 of the 15 SQL analyses produce validated outputs; target all 15. |
| Geography | Connecticut county/ZIP-level analysis is complete. |
| Network | At least one selected-plan network case study is validated or explicitly deferred due to insufficient public matching. |
| Story | At least 5 charts/maps support one coherent narrative. |
| Caveats | Every access claim distinguishes direct measurement from proxy. |

## 16. Sources and implementation notes

Official CMS Provider Data Catalog: https://developer.cms.gov/provider-data/ — machine-readable Medicare provider datasets, including hospitals and clinicians. cite turn0search3

NPPES downloadable files: https://download.cms.gov/nppes/NPI_Files.html — current V2 files and reference files. cite turn3search0

CMS Doctors & Clinicians: https://data.cms.gov/provider-data/topics/doctors-clinicians — National Downloadable File, Facility Affiliation, and Utilization datasets. cite turn1search0

CMS Hospital General Information: https://data.cms.gov/provider-data/dataset/xubh-q36u — current hospital general information dataset/API. cite turn0search6

CMS Medicare Physician & Other Practitioners: https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners — provider, service and geography products. cite turn2search12

CMS Marketplace Exchange PUFs: https://www.cms.gov/marketplace/resources/data/public-use-files — Plan, Service Area, Network and machine-readable URL files. cite turn2search1

CMS Transparency in Coverage: https://www.cms.gov/priorities/healthplan-price-transparency/overview/use-pricing-information-published-under-transparency-coverage-final-rule — machine-readable in-network/out-of-network files and implementation context. cite turn2search10

Census ACS and TIGER/Line: https://www.census.gov/data/developers/data-sets/acs-5year.html and https://www.census.gov/geographies/mapping-files/2025/geo/tiger-line-file.html. cite turn0search16 turn2search2

## 17. Recommended first build

Do not download every possible healthcare dataset on day one. Build the smallest end-to-end slice first:

13. Download the current NPPES V2 monthly file and reference files.
14. Filter to Connecticut for the first analytical run after raw preservation, while retaining the national raw source if storage permits.
15. Build unique NPI/provider and provider-location dimensions.
16. Load the CMS Doctors & Clinicians National Downloadable File and Facility Affiliation Data.
17. Load CMS Hospital General Information.
18. Load Census population/geography.
19. Produce provider density, specialty density, clinician-row-vs-NPI, provider-to-facility, and facility concentration analyses.
20. Only after those are stable, add Medicare utilization.
21. Then add Marketplace plan/network metadata.
22. Finally decide whether a selected-plan provider-level network match is strong enough to publish.

The first milestone should be a single reproducible chart: 'The healthcare provider funnel in Connecticut.' If that chart is analytically sound, the rest of the project has a strong foundation.